TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

AI.311

Robotics Planning

COURSE OVERVIEW

Deliberative planning for robots: task, motion, and manipulation planning under constraints.

1

SESSION

60 MINUTES

ORIENTATION: PLANNING TO ACT

KEY TOPICS

- Task vs. motion planning
- Why deliberate
- Planning in robotics
- Course roadmap

TEACHING FOCUS

Decide what to do before moving.

KEY TAKEAWAY Planning prevents costly mistakes.

ROBOTICS PLANNING

2

SESSION

60 MINUTES

TASK PLANNING

KEY TOPICS

- HTN and STRIPS planning
- Goal decomposition
- Plan execution and monitoring
- Replanning

TEACHING FOCUS

Structure complex missions.

KEY TAKEAWAY Task planning gives a robot purpose.

3

SESSION

60 MINUTES

MOTION PLANNING

KEY TOPICS

- Configuration space
- Roadmaps and PRM
- RRT and RRT*
- Optimal planning

TEACHING FOCUS

Find collision-free paths.

KEY TAKEAWAY Motion planning finds the way.

4

SESSION

60 MINUTES

MANIPULATION PLANNING

KEY TOPICS

- Grasp planning
- Pick-and-place
- Two-arm coordination
- Constrained manipulation

TEACHING FOCUS

Plan physical interaction with objects.

KEY TAKEAWAY Manipulation planning moves things intentionally.

5

SESSION

60 MINUTES

PLANNING UNDER UNCERTAINTY

KEY TOPICS

- Contingent planning
- Belief-space planning
- Online replanning
- Robust plans

TEACHING FOCUS

Plan when outcomes are uncertain.

KEY TAKEAWAY Good plans anticipate failure.

6

SESSION

60 MINUTES

ROBOTICS PLANNING & REVIEW

KEY TOPICS

- Planning demo on robot
- Course recap and assessment
- Toward social robotics (AI.312)
- Planning competition

TEACHING FOCUS

Execute a full task–motion plan.

KEY TAKEAWAY Plans turn goals into robot motion.